Redox mechanisms: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

All numerical cases are hypothetical, with the assumptions stated in each question. Give units and identify where a model would fail.

E01 Reaction and neutrality

For $O_O^{\times} \rightarrow \frac{1}{2}O_2 + V_O^{\bullet\bullet} + 2e'$, verify oxygen-atom count and effective charge. If electrons and doubly positive vacancies are the only charged species and local neutrality holds, find electron concentration when the vacancy concentration is $2 \times 10^{18}\,\mathrm{cm^{-3}}$. Name two reasons this estimate can fail.

E02 Drift and diffusion

In a dilute isothermal one-dimensional model, $J = -D\,dc/dx + (Dze/k_BT)cE$. Take $D = 10^{-18}\,\mathrm{m^2/s}$, $z = 2$, $E = 10^{8}\,\mathrm{V/m}$, $c = 10^{24}\,\mathrm{m^{-3}}$, $dc/dx = 10^{30}\,\mathrm{m^{-4}}$, and $k_BT = 4.14 \times 10^{-21}$ J. Use $e = 1.60 \times 10^{-19}$ C. Compute diffusion, drift and net number fluxes, including signs and units.

E03 Potential and migration barrier

A 12 nm film has a uniform 0.6 V drop. Calculate field magnitude, signed potential difference across a 0.2 nm hop along the electric field, and signed electrostatic endpoint energy difference for $z = +2$. Can that number alone determine the activation-barrier reduction? Explain.

E04 Gap sensitivity

In the hypothetical fixed-barrier model $I(g) = I_0 e^{-2\kappa g}$, $\kappa = 5\,\mathrm{nm^{-1}}$. Find $I(1.3\,\mathrm{nm})/I(1.0\,\mathrm{nm})$. Give two effects ignored by the model.

E05 Thermal transient

A lumped device starts at ambient with constant $P = 0.1$ mW, $R_{\mathrm{th}} = 10^{6}$ K/W and $C_{\mathrm{th}} = 10^{-13}$ J/K. Calculate the thermal time constant, steady-state temperature rise and temperature rise at $t = 100$ ns. Why is energy alone insufficient to find the peak temperature?

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Redox mechanisms: Exercises (continued)

E06 Activated timescale

Assume one thermally activated process at constant 300 K obeys $\tau = 10^{-12}\exp(E_a/k_BT)$ s, with $E_a = 1.0$ eV and $k_BT = 0.02585$ eV. Estimate τ in seconds and hours. What additional information is needed before claiming a retention time for a real memory?

E07 Mechanisms and boundaries

An Ag-based cell exhibits switching. Write an oxidation and a reduction half-reaction and identify where electrons and mobile ions may enter your explanation. Does this information determine the direction from which a filament grows? Contrast a VCM alternative and name two measurements that can help distinguish models.

E08 Hysteresis versus retention

A triangular voltage sweep produces an I–V loop passing through the origin. Design a controlled experiment that tests nonvolatile retention and explain why the crossing alone is insufficient. Name two possible causes of a nonzero current at zero applied terminal voltage.

E09 Boundary flux and threshold

An idealized $L = 10$ nm slab has a spatially averaged defect concentration $N(0) = 2 \times 10^{18}\,\mathrm{cm}^{-3}$. Assume constant injection and removal fluxes $s = 8 \times 10^{12}$ and $r = 2 \times 10^{12}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$, with no volume reactions. When does the average reach a model threshold $N_c = 5 \times 10^{18}\,\mathrm{cm}^{-3}$? Does this establish a local metallic filament?

E10 Compare same-polarity scan energy

Two positive voltage ramps start at zero with $dV/dt = 1$ V/s. SET has $I(V) = 1\,\mu\mathrm{S} \times V$ until 1 V; RESET has $I(V) = 4\,\mu\mathrm{S} \times V$ until 0.5 V. Integrate the energy delivered in each ramp, and explain whether their peak temperatures follow from the energy comparison alone.

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Redox mechanisms: Exercises (continued)

E11 Local concentration and threshold interval

A one-dimensional region of thickness $L = 10$ nm has fixed $c(0) = 8 \times 10^{18}$ and $c(L) = 2 \times 10^{18}\,\mathrm{cm}^{-3}$. Assume constant temperature, $D = 10^{-14}\,\mathrm{cm}^2/\mathrm{s}$, no drift or volume sources, and steady state. Use $J = -D dc/dx$ to find $c(x)$ and the flux magnitude and direction. If $N_c = 5 \times 10^{18}\,\mathrm{cm}^{-3}$, find the interval and length where $c(x) \geq N_c$. Does this establish a connected filament?

E12 Acceptor compensation

Suppose $[V_O^{\bullet\bullet}] = 3 \times 10^{18}\,\mathrm{cm}^{-3}$ and the concentration of singly negative ionized acceptors is $2 \times 10^{18}\,\mathrm{cm}^{-3}$. Assume local neutrality and no other charged species. Find the electron concentration, compare the result obtained if acceptors were omitted, and state where local neutrality may fail.

E13 Two prescribed moving fronts

A $L = 10$ nm film initially has $c_{\mathrm{low}} = 2 \times 10^{18}\,\mathrm{cm}^{-3}$ everywhere. In a purely illustrative sharp-front approximation, the high region has uniform $c_{\mathrm{high}} = 8 \times 10^{18}\,\mathrm{cm}^{-3}$, and the chosen threshold is $c_* = 5 \times 10^{18}\,\mathrm{cm}^{-3}$. In case A the high region begins at $x = 0$ and extends into the film at 2 nm/s; in case B it begins at $x = L$ and extends at 1 nm/s. Find each above-threshold interval and length at $t = 2$ s, and the net influx per unit area needed to sustain each front, assuming negligible flux through the opposite boundary and no volume source. For a further 2 s, assume each high region remains anchored at its original supply interface, its far front recedes toward that interface at 1 nm/s, and internal redistribution maintains this shape while the corresponding loss exits that interface. Find the final intervals and the outward flux magnitude while the front exists. Can net flux alone determine this geometry? Are the assigned directions universal consequences of terminal polarity?

Redox mechanisms: Exercises (continued)

E14 Boundary flux and inventory

A slab is 10 nm thick, has no bulk sources, and has boundary number fluxes defined positive along $+x$. During 2 s, suppose $J(0) - J(L) = 10^{12}\,\mathrm{cm^{-2}s^{-1}}$. Find the change in its spatially averaged concentration. What changes if both boundaries are blocking, and can the local profile still change? Compare environmental exchange, grain-boundary storage, reactive layers and approximately blocking electrodes; give one reason a nominally blocking electrode might not be a perfect boundary. Express a possible linear exchange law at $x = 0$ using an interfacial rate constant k and equilibrium concentration c_{eq}, and explain what further information is needed to solve the profile.

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